Part 1:

1. $\hat{S}_z = \frac{1}{2} \sum_{p\sigma} \sigma \alpha^{\dagger}_{p\sigma} \alpha_{p\sigma}$

$\hat{S}^2 = \hat{S}_z^2 + \frac{1}{2}\left(\hat{S}_+ S_- + \hat{S}_- \hat{S}_+\right)$

$\hat{S}_{\pm} = \sum_p \alpha^{\dagger}_{p\pm} \alpha_{p\mp}$

$\hat{H}_0 = \xi \sum_{p\sigma} (p-1) \alpha^{\dagger}_{p\sigma} \alpha_{p\sigma} \quad (\xi = 1)$

$\hat{V} = \frac{1}{2} g \sum_{pq} a^{\dagger}_{p+} a_{p-} a_{q-} a_{q+}$

(1) $[\hat{H}_0, \hat{S}_z] = 0$

$\hat{H}_0 \hat{S}_z = \hat{S}_z \hat{H}_0$ (Prove this)

$$\hat{H}_0 \hat{S}_z = \left[\sum_{p\sigma} (p-1) \alpha^\dagger_{p\sigma} \alpha_{p\sigma} \right] \times \left[\frac{1}{2} \sum_{p\sigma} \sigma \alpha^\dagger_{p\sigma} \alpha_{p\sigma} \right]$$

$$= \frac{1}{2} \left[\sum_{\substack{p\sigma \\ p'\sigma'}} \sigma' (p-1) \alpha^\dagger_{p\sigma} \alpha_{p\sigma} \alpha^\dagger_{p'\sigma'} \alpha_{p'\sigma'} \right]$$

For fermions

$$\left[\alpha_i^\dagger \alpha_i \, , \, \alpha_j^\dagger \alpha_j \right] = 0$$

$$= \frac{1}{2} \left[\sum_{\substack{p\sigma \\ p'\sigma'}} \sigma' (p-1) \alpha^\dagger_{p'\sigma'} \alpha_{p'\sigma'} \alpha^\dagger_{p\sigma} \alpha_{p\sigma} \right]$$

$$= \left[\frac{1}{2} \sum_{p'\sigma'} \sigma' \alpha^\dagger_{p'\sigma'} \alpha_{p'\sigma'} \right] \times \left[\sum_{p\sigma} (p-1) \alpha^\dagger_{p\sigma} \alpha_{p\sigma} \right]$$

$$= \hat{S}_z \hat{H}_0$$

(2) $\left[H_0 \, , \, \hat{S}_\pm \right] = 0$

$$\hat{H}_0 \hat{S}_\pm = \left[\sum_{p\sigma} (p-1) \alpha^\dagger_{p\sigma} \alpha_{p\sigma} \right] \times \left[\sum_{p} \alpha^\dagger_{p\pm} \alpha_{p\mp} \right]$$

$$= \sum_{p\sigma} (p-1) \alpha^\dagger_{p\sigma} \alpha_{p\sigma} \alpha^\dagger_{p'\pm} \alpha_{p'\mp}$$

$$= \sum_{\substack{p\sigma \\ p'}}^{p'} (p-1)\, a^{\dagger}_{p'\pm} a_{p'\mp}\, a^{\dagger}_{p\sigma} a_{p\sigma}$$

$$= \left[\sum_{p'} a_{p'\pm} a_{p'\mp} \right] \times \left[\sum_{p\sigma} (p-1)\, a^{\dagger}_{p\sigma} a_{p\sigma} \right]$$

$$= \hat{S}_{\pm} \hat{H}_0$$

$$\left[\hat{H}_0 , \hat{S}^2 \right] = 0$$

$$\hat{H}_0 \hat{S}^2 = \hat{S}^2 \hat{H}_0 \quad \text{(Prove this)}$$

$$\hat{H}_0 \hat{S}^2 = H_0 \left[\hat{S}_z^2 + \frac{1}{2}\left(\hat{S}_+ \hat{S}_- + \hat{S}_- \hat{S}_+ \right) \right]$$

$$= H_0 \hat{S}_z \hat{S}_z + \frac{1}{2} \hat{H}_0 \hat{S}_+ \hat{S}_- + \frac{1}{2} \hat{H}_0 \hat{S}_- \hat{S}_+$$

$$= \hat{S}_z^2 \hat{H}_0 + \frac{1}{2} \hat{S}_+ \hat{S}_- \hat{H}_0 + \frac{1}{2} \hat{S}_- \hat{S}_+ \hat{H}_0$$

$$= \left[\hat{S}_z^2 + \frac{1}{2}\left(\hat{S}_+ \hat{S}_- + \hat{S}_- \hat{S}_+ \right) \right] \hat{H}_0$$

$$= \underline{\underline{\hat{S}^2 \hat{H}_0}}$$

(3) $[\hat{V}, \hat{S}_z] = 0$

$\hat{V}\hat{S}_z = \hat{S}_z\hat{V}$ (prove this)

$[\hat{S}_z, \hat{P}_p^{\pm}] = 0$

$\hat{S}_z\hat{P}_p^{+} = \hat{P}_p^{+}\hat{S}_z$ (prove this first)

$$\hat{S}_z\hat{P}_p^{+} = \frac{1}{2}\sum_{p\sigma}\sigma\, a^{\dagger}_{p\sigma}a_{p\sigma}\, a^{\dagger}_{p'+}a^{\dagger}_{p'-}$$

$$\left[a^{\dagger}_{p\sigma}a_{p\sigma}\, a^{\dagger}_{p'+}a^{\dagger}_{p'-}\right] =$$

$$\left[a^{\dagger}_{p\sigma}\left(\delta_{pp'}\delta_{\sigma+} - a^{\dagger}_{p'+}a_{p\sigma}\right)a^{\dagger}_{p'-}\right] =$$

$$\left[a^{\dagger}_{p\sigma}a^{\dagger}_{p'-}\delta_{pp'}\delta_{\sigma+} - a^{\dagger}_{p\sigma}a^{\dagger}_{p'+}\left(\delta_{pp'}\delta_{\sigma-} - a^{\dagger}_{p'-}a_{p\sigma}\right)\right] =$$

$$\left[a^{\dagger}_{p\sigma}a^{\dagger}_{p'-}\delta_{pp'}\delta_{\sigma+} - a^{\dagger}_{p\sigma}a^{\dagger}_{p'+}\delta_{pp'}\delta_{\sigma-}\right.$$

$$\left. + a^{\dagger}_{p\sigma}a^{\dagger}_{p'-}a^{\dagger}_{p'+}a_{p\sigma}\right] -$$

$$\alpha_{p\sigma}\alpha_{p'+}\alpha_{p'-}\alpha_{p\sigma}\Big] =$$

$$\Big[\underbrace{\alpha^{\dagger}_{p\sigma}\alpha^{\dagger}_{p'-}\delta_{pp'}\delta_{\sigma+} - \alpha^{\dagger}_{p\sigma}\alpha^{\dagger}_{p'+}\delta_{pp'}\delta_{\sigma-}}_{\text{Non-zero only when } p = p'} + \alpha^{\dagger}_{p'+}\alpha^{\dagger}_{p'-}\alpha^{\dagger}_{p\sigma}\alpha_{p\sigma}\Big]$$

$$\Big[\underbrace{\alpha^{\dagger}_{p'\sigma}\alpha^{\dagger}_{p'-}\delta_{\sigma+} - \alpha^{\dagger}_{p'\sigma}\alpha^{\dagger}_{p'+}\delta_{\sigma-}}_{\text{Put back into the sum.}} + \alpha^{\dagger}_{p'+}\alpha^{\dagger}_{p'-}\alpha^{\dagger}_{p\sigma}\alpha_{p\sigma}\Big]$$

$$\frac{1}{2}\sum_{\sigma}\sigma\Big[\alpha^{\dagger}_{p'\sigma}\alpha^{\dagger}_{p'-}\delta_{\sigma+} - \alpha_{p'\sigma}\alpha_{p'+}\delta_{\sigma-}\Big] =$$

$$\frac{1}{2}\Big[\alpha^{\dagger}_{p'+}\alpha^{\dagger}_{p'-} + \alpha^{\dagger}_{p'-}\alpha^{\dagger}_{p'+}\Big] =$$

$$1\Big(\qquad \Big) \qquad \sim \ / \text{Anti-} \ \backslash$$

$$\frac{1}{2}\{\alpha_{p'+}, \alpha_{p'-}\} = 0 \quad \text{(commutation)}$$

This means that:

$$\hat{S}_z \hat{P}_p^+ = \frac{1}{2}\sum_{p\sigma} \sigma \, \alpha^+_{p\sigma} \alpha_{p\sigma} \alpha^+_{p'+} \alpha^+_{p'-}$$

$$= \frac{1}{2}\sum_{p\sigma} \sigma \, \alpha^+_{p'+} \alpha^+_{p'-} \alpha^+_{p\sigma} \alpha_{p\sigma}$$

$$= \underline{\hat{P}_p^+ \hat{S}_z}$$

$$[\hat{S}_z, \hat{P}_p^+] = 0$$

$$[\hat{S}_z, \hat{P}_p^-] = 0 \quad \text{(can be shown in the same way)}$$

$$\hat{V} = -\frac{1}{2} g \sum_{pq} \hat{P}_p^+ \hat{P}_q^-$$

$$\hat{V}\hat{S}_z = -\frac{1}{2}g\sum_{pq}\left[\hat{P}_p^+\hat{P}_q^-\right]\hat{S}_z$$

$$= -\frac{1}{2}g\sum_{pq}\hat{S}_z\left[\hat{P}_p^+\hat{P}_q^-\right]$$

$$= \hat{S}_z\left[-\frac{1}{2}g\sum_{pq}\left[\hat{P}_p^+\hat{P}_q^-\right]\right]$$

$$= \underline{\hat{S}_z\hat{V}}$$

(4) $\left[\hat{V}, \hat{S}^2\right] = 0$

$\hat{V}\hat{S}^2 = \hat{S}^2\hat{V}$ (Prove this)

$\left[\hat{S}_\pm, \hat{P}_p^\pm\right] = 0$

$$\hat{S}_{\pm}\hat{P}_p^{+} = \hat{P}_p^{+}\hat{S}_{\pm} \quad \text{(Prove this first)}$$

$$\hat{S}_{\pm}\hat{P}_p^{+} = \sum_p \alpha^{\dagger}_{p\pm}\alpha_{p\mp}\alpha^{\dagger}_{p'+}\alpha^{\dagger}_{p'-}$$

$$\left[\alpha^{\dagger}_{p\pm}\alpha_{p\mp}\alpha^{\dagger}_{p'+}\alpha^{\dagger}_{p'-}\right] =$$

$$\left[\alpha^{\dagger}_{p\pm}\left(\delta_{pp'}\delta_{\mp+} - \alpha^{\dagger}_{p'+}\alpha_{p\mp}\right)\alpha^{\dagger}_{p'-}\right] =$$

$$\left[\alpha^{\dagger}_{p\pm}\alpha^{\dagger}_{p'-}\delta_{pp'}\delta_{\mp+} - \alpha^{\dagger}_{p\pm}\alpha^{\dagger}_{p'+}\left(\delta_{pp'}\delta_{\mp-} - \alpha^{\dagger}_{p'-}\alpha_{p\mp}\right)\right] =$$

$$\left[\alpha^{\dagger}_{p\pm}\alpha^{\dagger}_{p'-}\delta_{pp'}\delta_{\mp+} - \alpha^{\dagger}_{p\pm}\alpha^{\dagger}_{p'+}\delta_{pp'}\delta_{\mp-} + \alpha^{\dagger}_{p\pm}\alpha^{\dagger}_{p'+}\alpha^{\dagger}_{p'-}\alpha_{p\mp}\right] =$$

$$\left[\underbrace{\alpha^{\dagger}_{p'\pm}\alpha^{\dagger}_{p'-}\delta_{\mp+} - \alpha^{\dagger}_{p'\pm}\alpha^{\dagger}_{p'+}\delta_{\mp-}}_{\downarrow} + \alpha^{\dagger}_{p\pm}\alpha^{\dagger}_{p'+}\alpha^{\dagger}_{p'-}\alpha_{p\mp}\right] =$$

Both are always 0, since we either get $\delta_{\pm/\mp} = 0$ or $\alpha^{\dagger}\alpha^{\dagger}$

$$= \left[\alpha^{\dagger}_{p'+} \alpha^{\dagger}_{p'-} \alpha^{\dagger}_{p'\pm} \alpha_{p'\mp} \right]$$

$$\hat{S}_{\pm} \hat{P}^{+}_{p} = \sum_{p} \alpha^{\dagger}_{p\pm} \alpha_{p\mp} \alpha^{\dagger}_{p'+} \alpha^{\dagger}_{p'-}$$

$$= \alpha^{\dagger}_{p'+} \alpha^{\dagger}_{p'-} \sum_{p} \alpha^{\dagger}_{p\pm} \alpha_{p\mp}$$

$$= \underline{\hat{P}^{+}_{p} \hat{S}_{\pm}}$$

$$\left[\hat{S}_{\pm}, \hat{P}^{+}_{p} \right] = 0$$

$$\left[\hat{S}_{\pm}, \hat{P}^{-}_{p} \right] = 0 \quad \left(\begin{array}{l} \text{Can be shown} \\ \text{in the same way} \end{array} \right)$$

$$\hat{V}\hat{S}^{2} = \frac{1}{\quad} \left[\sum \hat{P}^{+} \hat{P}^{-} \right] \left[\hat{S}^{2} \right]$$

$$2\,g\left[\langle pq|\,p\,|q\rangle\right]\times\left[S_z+\frac{1}{2}\left(S_+S_-+S_-S_+\right)\right]$$

$$=\left[\hat{S}_z^2+\frac{1}{2}\left(\hat{S}_+\hat{S}_-+\hat{S}_-\hat{S}_+\right)\right]\times\left[-\frac{1}{2}g\sum_{pq}\hat{P}_p^+\hat{P}_q^-\right]$$

$$=\hat{S}^2\hat{V}$$

(5) $$\hat{H}=\xi\sum_{p\sigma}(p-1)\,a^+_{p\sigma}a_{p\sigma}$$

$$-\frac{1}{2}g\sum_{pq}a^+_{p+}a^+_{p-}a_{q-}a_{q+}$$

$$\hat{P}_p^+=a^+_{p+}a^+_{p-}$$

$$\hat{P}_q^-=a^+_{q-}a^+_{q+}$$

$$\xi=1$$

$$\hat{H}=\sum_{p\sigma}(p-1)\,a^+_{p\sigma}a_{p\sigma}-\frac{1}{2}\sum_{pq}\hat{P}_p^+\hat{P}_q^-$$

2. $\Phi_{pq}(x_1, x_2, x_3, x_4) =$

$$\begin{vmatrix} \psi_{p+}(x_1) & \psi_{p+}(x_2) & \psi_{p+}(x_3) & \psi_{p+}(x_4) \\ \psi_{p-}(x_1) & \psi_{p-}(x_2) & \psi_{p-}(x_3) & \psi_{p-}(x_4) \\ \psi_{q+}(x_1) & \psi_{q+}(x_2) & \psi_{q+}(x_3) & \psi_{q+}(x_4) \\ \psi_{q-}(x_1) & \psi_{q-}(x_2) & \psi_{q-}(x_3) & \psi_{q-}(x_4) \end{vmatrix}$$

$$|\alpha\beta\rangle = \Phi_{\alpha\beta}(x_1, x_2, x_3, x_4)$$

$$\langle\phi|\hat{H}|\phi\rangle = \sum_{p\sigma}(p-1)\langle\phi|\overbrace{a^{\dagger}_{p\sigma}a_{p\sigma}}|\phi\rangle$$

$$- \frac{1}{2}g\sum_{pq}\langle\phi|\overbrace{a^{\dagger}_{p+}\underbrace{a^{\dagger}_{p-}a_{q-}}a_{q+}}|\phi\rangle$$

$$= \sum_{p\sigma}(p-1)\langle\phi|\phi\rangle - \frac{1}{2}g\sum_{pq}\langle\phi|\phi\rangle\delta_{pq}$$

$$= 2\sum_{p}(p-1) - \frac{1}{2}g\sum_{pq}\delta_{pq}$$

$$= 2(\alpha - 1) + 2(\beta - 1) - \frac{1}{2} g \cdot 2$$

$$= \underline{2(\alpha + \beta - 2) - g}$$

$$\langle \Phi | \hat{H} | \Phi_i^a \rangle = \langle \Phi | \hat{H} \, a_{a+}^\dagger a_{a-}^\dagger a_{i-} a_{i+} | \Phi \rangle$$

$$= \sum_{p\sigma} (p-1) \langle \Phi | a_{p\sigma}^\dagger a_{p\sigma} a_{a+}^\dagger a_{a-}^\dagger a_{i-} a_{i+} | \Phi \rangle$$

$$- \frac{1}{2} g \sum_{pq} \langle \Phi | a_{p+}^\dagger a_{p-}^\dagger a_{q-} a_{q+}^\dagger a_{a+}^\dagger a_{a-}^\dagger a_{i-} a_{i+} | \Phi \rangle$$

$$= \sum_{p\sigma} (p-1) \langle \Phi | \Phi \rangle \cdot 0 - \frac{1}{2} g \sum_{pq} \langle \Phi | \Phi \rangle \, \delta_{pi} \, \delta_{qa}$$

$$= 0 - \frac{1}{2} g \cdot 1$$

$$= \underline{-\frac{1}{2} g}$$

$$\langle \Phi | \hat{H} | \Phi_{ij}^{ab} \rangle = 0$$

(6) $[\hat{P}_p^+ , \hat{P}_q^+] = 0$

$$\hat{P}_p^+ \hat{P}_q^+ = a_{p+}^+ a_{p-}^+ a_{q+}^+ a_{q-}^+$$

$$= a_{p+}^+ a_{q+}^+ a_{q-}^+ a_{p-}^+$$

$$= a_{q+}^+ a_{q-}^+ a_{p+}^+ a_{p-}^+$$

$$= \hat{P}_q^+ \hat{P}_p$$

Same math for showing:

$[\hat{P}_p^- , \hat{P}_q^-] = 0$

Part 3:

5) $\hat{H} = \sum_{p=1}^{L} \varepsilon_p \hat{n}_p - \frac{g}{2} \sum_{pq} \hat{P}_p^+ \hat{P}_p^-$

$\varepsilon_p = \varepsilon(p-1)$

ε_p is the energi of site p and $\hat{n}_p$ checks wether site p is occupied. $\sum_{p=1}^{L} \varepsilon_p \hat{n}_p \rightarrow$ sum of single particle energies of all states occupied in $|n_1 n_2 \dots n_L\rangle$.

$P_p^+ P_q^-$ removes pair at positon q and creates a pair at position p.

g is the pairing interaction streangth.

$-\frac{g}{2} \sum_{pq} P_p^+ P_q^-$ sums over all pairs,

where pair at site q can be annihalet and at site p can be created. Eeach contributing with $-\frac{J}{2}$.

$$\{|0\rangle, |1\rangle\} \rightarrow |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\left.\begin{aligned} \hat{p}_p^+ |0\rangle_p &= |1\rangle_p \\ \hat{p}_p^+ |1\rangle_p &= \vec{0}_p \end{aligned}\right\} \quad \hat{p}_p^+ = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\left.\begin{aligned} \hat{p}_p^- |1\rangle_p &= |0\rangle_p \\ \hat{p}_p^- |0\rangle_p &= \vec{0}_p \end{aligned}\right\} \quad \hat{p}_p^- = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\hat{n}_p = \hat{n}_{p+} + \hat{n}_{p-}$$

$$\left.\begin{aligned} \hat{n}_p |0\rangle_p &= 0|0\rangle_p \\ \hat{n}_p &= \end{aligned}\right\} \quad \hat{n}_p = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$\ln p\,|1\rangle_p = 2\,|1\rangle_p$)

States of the Automaton:

1. Every site so far did nothing.
2. $\hat{p}^+$ applied to this site or previous. some site to the right must apply $\hat{p}^-$ (start term).
3. $\hat{p}^-$ applied to this site or previous. som site to the right must apply $\hat{p}^+$ (start term).
4. Term is done, do nothing from here.

$D = 4$ since between two

neighbouring sites, the left site only need 1 of 4 possible "messages". this is independant of L since "messages" never record which site started a term.

$W^{[p]}$ = 4 x 4 Matrix where element (α, β) is the operator applied when going from state of automaton α to state of automaton β.

$$W^{[p]} = \begin{bmatrix} I & P_p^+ & P_p^- & \varepsilon_p \Lambda_p - \frac{g}{2} P_p^+ P_p^- \\ 0 & I & 0 & -\frac{g}{2} P_p^- \\ 0 & 0 & I & -\frac{g}{2} P_p^+ \\ 0 & 0 & 0 & I \end{bmatrix}$$

$$W^{[1]} = \begin{bmatrix} I & P_p^+ & P_p^- & \underbrace{\varepsilon_p n_p - \frac{g}{2} P_p^+ P_p^-}_{\downarrow} \end{bmatrix}$$

$$W^{[4]} = \begin{bmatrix} \varepsilon_p n_p - \frac{g}{2} P_p^+ P_p^- \\ -\frac{g}{2} P_p^- \\ -\frac{g}{2} P_p^+ \\ I \end{bmatrix} \quad X_p$$

$$W^{[1]} \cdot W^{[2]} = \begin{bmatrix} I & P_2^+ & P_2^- & X_2 \\ 0 & I & 0 & -\frac{g}{2} P_2^- \\ 0 & 0 & I & -\frac{g}{2} P_2^+ \\ 0 & 0 & 0 & I \end{bmatrix}$$

$$\begin{bmatrix} I & P_1^+ & P_1^- & X_1 \end{bmatrix}$$

$$\begin{bmatrix} I, P_1^+ + P_2^+, P_1^- + P_2^-, X_2 - \frac{g}{2} P_1^+ P_2^- \\ -\frac{g}{2} P_1^- P_2^+ + X_1 \end{bmatrix}$$

$$W^{[1]} W^{[2]} W^{[3]} = \Big[I, \; p_1^+ + p_2^+ + p_3^+, \; p_1^- + p_2^- + p_3^-$$

$$X_3 - \frac{g}{2} p_1^+ p_3^- - \frac{g}{2} p_2^+ p_3^- - \frac{g}{2} p_1^- p_3^+ - \frac{g}{2} p_1^- p_3^+$$

$$+ X_2 - \frac{g}{2} p_1^+ p_2^- - \frac{g}{2} p_1^- p_2^+ + X_1 \Big]$$

$$W^{[1]} W^{[2]} W^{[3]} W^{[4]} = X_4 + X_3$$

$$+ X_2 + X_1 - \frac{g}{2} \Big[p_1^+ p_4^- + p_2^+ p_4^- + p_3^+ p_4^- +$$

$$p_4^+ p_1^- + p_4^+ p_2^- + p_4^+ p_3^- + p_1^+ p_3^- + p_2^+ p_3^- +$$

$$p_3^+ p_1^- + p_3^+ p_2^- + p_1^+ p_2^- + p_2^+ p_1^- \Big] =$$

$$\sum_{p=1}^{4} \varepsilon_p n_p - \frac{g}{2} \sum_{pq} p_p^+ p_q^- = \hat{H}$$

Constructing the hamiltonian:

1. $\varepsilon_p n_p$ gives ε_p if p is occupied
2. $-\frac{g}{2}[P_p^+ P_q^-]$ gives $-\frac{g}{2}$ if $p=q$ and p is occupied.
3. $-\frac{g}{2}[P_p^+ P_q^-]$ $(p \neq q)$ removes pair at q and adds to p. This gives 0 if p is unnoccupied or q is occupied.

$$H|11007 = \left[\varepsilon_1 + \varepsilon_2 - g\right]|1100\rangle - \frac{g}{2}\left[|1010\rangle + |0110\rangle + |1001\rangle + |0101\rangle\right] + 0|0011\rangle$$

$$H|1010\rangle = [\varepsilon_1 + \varepsilon_3 - g]|1010\rangle$$
$$- \frac{g}{2}\Big[|1100\rangle + |0110\rangle + |1001\rangle + |0011\rangle\Big] + 0|0101\rangle$$

$$H|0110\rangle = [\varepsilon_2 + \varepsilon_3 - g]|0110\rangle$$
$$- \frac{g}{2}\Big[|1100\rangle + |1010\rangle + |0101\rangle + |0011\rangle\Big] + 0\cdot|1001\rangle$$

$$H|1001\rangle = [\varepsilon_1 + \varepsilon_4 - g]|1001\rangle$$
$$- \frac{g}{2}\Big[|1100\rangle + |1010\rangle + |0101\rangle + |0011\rangle\Big] + 0\cdot|0110\rangle$$

$$H|0101\rangle = [\varepsilon_2 + \varepsilon_4 - g]|0101\rangle$$
$$- g($$

$$\frac{v}{2}\left[|1100\rangle + |0110\rangle + |1001\rangle + |0011\rangle\right] + 0\cdot|1010\rangle$$

$$H|0011\rangle = [\varepsilon_3 + \varepsilon_4 - J]|0011\rangle - \frac{J}{2}\left[|1010\rangle + |0110\rangle + |1001\rangle + |0101\rangle\right] + 0\cdot|1100\rangle$$

$$H = \begin{array}{c|cccccc} & |1100\rangle & |1010\rangle & |0110\rangle & |1001\rangle & |0101\rangle & |0011\rangle \\ \hline |1100\rangle & \varepsilon_1+\varepsilon_2-J & -\frac{J}{2} & -\frac{J}{2} & 0 & 0 & 0 \\ |1010\rangle & -\frac{J}{2} & \varepsilon_1+\varepsilon_3-J & 0 & -\frac{J}{2} & 0 & 0 \\ |0110\rangle & -\frac{J}{2} & 0 & \varepsilon_2+\varepsilon_3-J & 0 & -\frac{J}{2} & 0 \\ |1001\rangle & 0 & -\frac{J}{2} & 0 & \varepsilon_1+\varepsilon_4-J & 0 & 0 \\ |0101\rangle & 0 & 0 & -\frac{J}{2} & 0 & \varepsilon_2+\varepsilon_4-J & 0 \\ |0011\rangle & 0 & 0 & 0 & 0 & 0 & \varepsilon_3+\varepsilon_4-J \end{array}$$

Gives same H as in exercise 2 if $\epsilon = 1$.

if we increase d from $d=2$ to $d=4$, D will still be $D=4$. Size of W stays the same $D \times D$, but size of I, P_0^+, P_0^-, n_p must increase from 2×2 to 4×4.

6. In MPOs operators on different sites must commute. For fermions, the hamiltonian often has operators that anti-comutes, like $\{a_p, a_q\}$, $\{a_p^\dagger, a_q^\dagger\}$. to force these operations to

commute we add Jordan-Wigner-transformations on sites between that make the operators commute. These chains of Jordan-Wigner-transformations are called a Parity string.

Luckily our hamiltonian only has opperators that commute as shown in exercise 1, so we avoid this problem. it dosnt matter if $d=2$ or $d=4$, our hamiltonian dosnt allow single fermion excitations and the opperators commute.

if we do allow single-particle excitations (for example $a^\dagger$ ...)

excitations (for example $a_{p\sigma} a_{p\sigma}$) we must make sure the operation commutes between sites. This is done by adding jordan winger transformation that creates a parity string between the operators. These transformation appear in the waiting entries of the MPO.

for our $W^{[p]}$ it would look something like:

$$\begin{bmatrix} I & P_p^+ & P_p^- & \epsilon_p n_p - (g/2) P_p^+ P_p^- \\ 0 & F & 0 & -(g/2) P_p^- \\ 0 & 0 & F & -(g/2) P_p^+ \\ 0 & 0 & 0 & I \end{bmatrix}$$

The F is parity matrix and

in this case we assume

$\{p_p^+, p_q^+\}$, $\{p_p^-, p_4^-\}$.